# Complex Numbers: Cartesian Form & Algebra

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Cartesian Arithmetic & Complex Conjugate**

Consider the complex numbers $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. Find the value of $z_1 - z_2$ in the form $x + yi$.

We are asked to find the difference between two complex numbers, $z_1$ and $z_2$. Given $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. We need to calculate $z_1 - z_2$. $z_1 - z_2 = (5 - 2i) - (1 + 4i)$ To subtract complex numbers, we subtract the real parts and the imaginary parts separately. Real part: $5 - 1 = 4$ Imaginary part: $(-2i) - (4i) = -2i - 4i = -6i$ Therefore, $z_1 - z_2 = 4 - 6i$.

**2** **[Foundational]** • **Target: Modulus and Argument**

Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \leq \pi$. Hence find $z^6$ and determine all three solutions to $w^3 = z$.

1. Modulus: $r = |z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$. Argument: since $z$ lies in Quadrant II, $\theta = \pi - \arctan(\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$. Polar form: $z = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$; Exponential form: $z = 2e^{i2\pi/3}$. 2. Compute $z^6$ via De Moivre's Theorem: $z^6 = 2^6 e^{i\cdot 6(2\pi/3)} = 64e^{i4\pi} = 64(\cos 4\pi + i\sin 4\pi) = 64(1+0i) = 64.$ 3. Solve $w^3 = z = 2e^{i2\pi/3}$.

**3** **[Foundational]** • **Target: Solving Complex Equations**

Consider the complex number $w$ such that $w = x + yi$, where $x$ and $y$ are real numbers. Given that $w^3 = -64i$ and the imaginary part of $w$ is positive, find the value of $x^2 + y^2$.

We are given that $w^3 = -64i$. We can express $-64i$ in polar form. The modulus is $|-64i| = 64$. The argument is $\arg(-64i) = \frac{3\pi}{2} + 2k\pi$ for integer $k$. Thus, $-64i = 64e^{i(\frac{3\pi}{2} + 2k\pi)}$. Let $w = re^{i\theta}$. Then $w^3 = r^3e^{i3\theta}$. Equating the moduli, we have $r^3 = 64$, so $r = 4$ (since $r$ must be real and non-negative). Equating the arguments, we have $3\theta = \frac{3\pi}{2} + 2k\pi$. Therefore, $\theta = \frac{\pi}{2} + \frac{2k\pi}{3}$. For $k = 0$.

**4** **[Foundational]** • **Target: Argand Diagram Geometry & Loci**

Let $z_1$ and $z_2$ be complex numbers such that $z_1 = 2 + i$ and $z_2 = 3 - 2i$. ; Consider the complex number $w = \frac{z_1^2}{z_2}$. ; If $w = a + bi$, where $a$ and $b$ are real numbers, find the value of $a^2 + b^2$.

First, we calculate $z_1^2$: $z_1^2 = (2+i)^2 = 4 + 4i + i^2 = 4 + 4i - 1 = 3 + 4i$. ; Next, we calculate $w = \frac{z_1^2}{z_2} = \frac{3+4i}{3-2i}$. ; To simplify, we multiply the numerator and denominator by the conjugate of the denominator: $w = \frac{(3+4i)(3+2i)}{(3-2i)(3+2i)} = \frac{9+6i+12i+8i^2}{9-4i^2} = \frac{9+18i-8}{9+4} = \frac{1+18i}{13} = \frac{1}{13} + \frac{18}{13}i$. ; So, $a = \frac{1}{13}$ and $b = \frac{18}{13}$. ; We need to find $a^2 + b^2$.

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**5** [Foundational] • Target: Cartesian Arithmetic & Complex Conjugate

The diagram shows the graph of the function $f(x) = x^2 - 4x + 3$ and a line $y = g(x)$. The line $y = g(x)$ intersects the parabola $y = f(x)$ at points A and B. Point A is given as $(1, 0)$. Find the x-coordinate of point B.

The points of intersection A and B are found by setting the two functions equal to each other: $f(x) = g(x)$. We are given $f(x) = x^2 - 4x + 3$ and from the diagram, the line is $y = x - 1$, so $g(x) = x - 1$. Setting them equal: $x^2 - 4x + 3 = x - 1$. Rearranging the equation to form a quadratic equation: $x^2 - 4x - x + 3 + 1 = 0$, which simplifies to $x^2 - 5x + 4 = 0$. We can factor this quadratic equation: $(x - 1)(x - 4) = 0$. The solutions for x are $x = 1$ and $x = 4$. We are given that point A has an x-c.

**6** [Foundational] • Target: Modulus and Argument

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(1, 4)$, and it passes through the point $(-1, 0)$. Find the value of $b$.

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where $(h, k)$ is the vertex. Given the vertex is $(1, 4)$, we have $f(x) = a(x - 1)^2 + 4$. The parabola passes through the point $(-1, 0)$. Substituting these coordinates into the equation: $0 = a(-1 - 1)^2 + 4$ $0 = a(-2)^2 + 4$ $0 = 4a + 4$ $4a = -4$ $a = -1$. So, the function is $f(x) = -1(x - 1)^2 + 4$. Expanding this: $f(x) = -(x^2 - 2x + 1) + 4$ $f(x) = -x^2 + 2x - 1 + 4$ $f(x) = -x^2 + 2x + 3$. Comparing this to the general form.

**7** [Foundational] • Target: Solving Complex Equations

For the rational function $f(x) = \frac{x^2 - x - 2}{x - 3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

(a) Vertical Asymptote: The denominator is zero at $x = 3$. Numerator at $x = 3$ is $3^2 - 3 - 2 = 4eq0$. Thus $x = 3$ is a vertical asymptote. Oblique Asymptote: Perform polynomial long division: $\frac{x^2 - x - 2}{x - 3} = x + 2 + \frac{4}{x - 3}$. As $x \to \pm\infty$, $\frac{4}{x - 3} \to 0$. Therefore, $y = x + 2$ is the oblique asymptote. (b) Differentiate $f(x) = x + 2 + 4(x - 3)^{-1}$: $f'(x) = 1 - \frac{4}{(x - 3)^2}$. Set $f'(x) = 0$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: Argand Diagram Geometry & Loci

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \le \pi$. Show that the sum of these five roots equals 0.

1. Write $-32 = 32e^{i\pi}$. By De Moivre's Theorem: $z_k = 32^{1/5}\exp\left(i\frac{\pi + 2\pi k}{5}\right) = 2e^{i(\pi + 2\pi k)/5}$ for $k = 0, 1, 2, 3, 4$. Evaluating the arguments in $(-\pi, \pi]$: - $k = 0$: $z_0 = 2e^{i\pi/5}$ - $k = 1$: $z_1 = 2e^{i3\pi/5}$ - $k = 2$: $z_2 = 2e^{i5\pi/5} = 2e^{i\pi} = -2$ - $k = 3$: $z_3 = 2e^{i7\pi/5} = 2e^{-i3\pi/5}$ - $k = 4$: $z_4 = 2e^{i9\pi/5} = 2e^{-i\pi/5}$ 2. Sum of the roots: Notice that.

**9** [Intermediate] • Target: Cartesian Arithmetic & Complex Conjugate

The diagram shows two similar triangles ABC and ADE, where D lies on AB and E lies on AC. It is given that AD = 6 cm, DB = 4 cm, and DE = 12 cm. Find the length of BC.

Since triangles ABC and ADE are similar, the ratio of corresponding sides is equal. The ratio of similarity between triangle ADE and triangle ABC can be found by comparing the lengths of corresponding sides AD and AB. We have AB = AD + DB = 6 + 4 = 10 cm. The ratio of similarity, k, is $\frac{AB}{AD} = \frac{10}{6} = \frac{5}{3}$. Alternatively, we can consider the ratio of similarity from ADE to ABC as $\frac{AD}{AB} = \frac{6}{10} = \frac{3}{5}$. Using this ratio, we can find the length of BC. The corresponding s.

**10** [Intermediate] • Target: Modulus and Argument

Find the Cartesian symmetric equation of the line of intersection between planes $\pi_1 : x + y + z = 1$ and $\pi_2 : 2x - y + 3z = 4$.

1. Normal vectors: $\mathbf{n}_1 = \langle 1, 1, 1\rangle$, $\mathbf{n}_2 = \langle 2, -1, 3\rangle$. 2. Direction vector $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$: $\mathbf{d} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, 1, 1; 2, -1, 3| = \mathbf{i}(3 - (-1)) - \mathbf{j}(3 - 2) + \mathbf{k}(-1 - 2) = (4; -1; -3)$. 3. Find a point on the line: set $z = 0$: $\{x + y = 1; 2x - y = 4 \Rightarrow 3x = 5 \Rightarrow x = \frac{5}{3},\ y = 1 - \frac{5}{3} = -\frac{2}{3}$. P.

**11** [Intermediate] • Target: Solving Complex Equations

Consider the complex number $z$ such that $z = a + 5i$ and $|z| = \sqrt{26}$. Given that $a$ is a real number, find the possible values of $a$.

The modulus of a complex number $z = x + yi$ is given by $|z| = \sqrt{x^2 + y^2}$. We are given $z = a + 5i$ and $|z| = \sqrt{26}$. Therefore, $|z| = \sqrt{a^2 + 5^2} = \sqrt{a^2 + 25}$. We are given that $|z| = \sqrt{26}$. So, we have the equation: $\sqrt{a^2 + 25} = \sqrt{26}$. Squaring both sides gives: $a^2 + 25 = 26$. Subtracting 25 from both sides, we get: $a^2 = 1$. Taking the square root of both sides, we find: $a = \pm 1$. Common mistakes include: ignoring the fact that $a$ can be negative, or incorrectly.

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**12** [Intermediate] • Target: Argand Diagram Geometry & Loci

Consider the points $A(1, 2, 3)$ and $B(4, 6, 9)$. Find the vector $\vec{AB}$ in component form.

To find the vector $\vec{AB}$, we subtract the coordinates of the initial point $A$ from the coordinates of the terminal point $B$. Let $A = (x_1, y_1, z_1) = (1, 2, 3)$ and $B = (x_2, y_2, z_2) = (4, 6, 9)$. The vector $\vec{AB}$ is given by $(x_2 - x_1; y_2 - y_1; z_2 - z_1)$. Substituting the given coordinates: $\vec{AB} = (4 - 1; 6 - 2; 9 - 3) = (3; 4; 6)$. Therefore, the vector $\vec{AB}$ in component form is $(3; 4; 6)$.

**13** [Intermediate] • Target: Cartesian Arithmetic & Complex Conjugate

The graph of the function $f(x) = \frac{x^2 - ax + 6}{x - 1}$ is shown below. The graph has a vertical asymptote at $x = 1$ and a point of discontinuity at $(2, k)$. Find the value of $a$.

The function is given by $f(x) = \frac{x^2 - ax + 6}{x - 1}$. $x = 1$ is a vertical asymptote, which means that $x - 1$ is not a factor of the numerator, and when $x = 1$, the numerator is non-zero. The point of discontinuity is at $(2, k)$. This implies that $x = 2$ is a root of the numerator, so $f(2) = 0$ or the expression is undefined at $x = 2$ if $x - 2$ were a factor of the denominator. Since the denominator is $x - 1$, which is not zero at $x = 2$, the function must be zero at $x = 2$ for there to be a removable discontinuity (a.

**14** [Intermediate] • Target: Modulus and Argument

Consider the function $f(x) = \frac{x^2 - 9}{x - k}$. The graph of $y = f(x)$ has a removable discontinuity at $x = 3$. Find the value of $k$. The graph of $y = f(x)$ is shown below.

For the function $f(x) = \frac{x^2 - 9}{x - k}$ to have a removable discontinuity at $x = 3$, the numerator and the denominator must both be zero at $x = 3$. This means that $(x - 3)$ must be a factor of both the numerator and the denominator. The numerator is $x^2 - 9$, which factors as $(x - 3)(x + 3)$. For the denominator to have a factor of $(x - 3)$, we must have $x - k = 0$ when $x = 3$. Thus, $3 - k = 0$, which implies $k = 3$. When $k = 3$, $f(x) = \frac{(x - 3)(x + 3)}{x - 3}$. For $xeq3$, $f(x) = x + 3$. The discontinuit.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Solving Complex Equations • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of $k$.

Since the parabola intersects the x-axis at $(2, 0)$ and $(6, 0)$, these are the roots of the quadratic equation $f(x) = 0$. Therefore, we can write the function in factored form as $f(x) = a(x - 2)(x - 6)$. Expanding this, we get $f(x) = a(x^2 - 8x + 12)$. Comparing this with the given function $f(x) = x^2 - 8x + k$, we can see that the coefficient of $x^2$ is 1. Thus, $a = 1$. So, $f(x) = x^2 - 8x + 12$. By comparing the constant terms, we find that $k = 12$. Alternatively, we can substitute one of the roots into the o.

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**16** **[Challenging]** • **Target: Argand Diagram Geometry & Loci** • **Cross-Lesson: 6.03 Markov Chains (AI HL)**

A continuous random variable $X$ has probability density function $f(x) = kx(2-x)$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere. Determine: (a) the constant $k$; (b) the median of $X$; (c) the variance $\mathrm{Var}(X)$.

1. Normalization condition: $\int_0^2 k(2x - x^2)\,dx = 1 \Rightarrow k\left[x^2 - \frac{x^3}{3}\right]_0^2 = k\left(4 - \frac{8}{3}\right) = k\left(\frac{4}{3}\right) = 1 \Rightarrow k = \frac{3}{4}$. 2. Median $m$: By visual symmetry of the parabola $f(x) = \frac{3}{4}x(2-x)$ about $x = 1$, $m = 1$. We verify: $F(1) = \int_0^1 \frac{3}{4}(2x - x^2)\,dx = \frac{3}{4}\left[x^2 - \frac{x^3}{3}\right]_0^1 = \frac{3}{4}\left(\frac{2}{3}\right) = \frac{1}{2}$. ✓ 3. Variance: By symmetry, $E[X] = 1$..

**17** **[Challenging]** • **Target: Cartesian Arithmetic & Complex Conjugate** • **Cross-Lesson: 1.01 Proof by Mathematical Induction**

The graph of the function $f(x) = x^3 - 6x^2 + 5$ is shown below. The graph has a local maximum and a local minimum. Find the x-coordinate of the local minimum.

To find the local minimum, we first need to find the derivative of the function $f(x)$. \\ \\ $f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 5) = 3x^2 - 12x$. \\ \\ Local extrema occur where $f'(x) = 0$. So, we set $3x^2 - 12x = 0$. \\ \\ Factoring out $3x$, we get $3x(x-4) = 0$. \\ \\ This gives us two critical points: $x = 0$ and $x = 4$. \\ \\ To determine whether these points are local maxima or minima, we can use the second derivative test. \\ \\ The second derivative is $f''(x) = \frac{d}{dx}(3x^2 - 12x) = 6x - 12$. \\.

**18** **[Challenging]** • **Target: Modulus and Argument** • **Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity**

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. What are the coordinates of point A?

To find the x-intercepts of the function $f(x) = x^2 - 6x + 8$, we need to solve for $x$ when $f(x) = 0$. This means we need to find the roots of the quadratic equation $x^2 - 6x + 8 = 0$. We can factor this quadratic equation. We are looking for two numbers that multiply to 8 and add to -6. These numbers are -2 and -4. Therefore, the equation can be factored as $(x-2)(x-4) = 0$. Setting each factor to zero gives us the x-intercepts: $x - 2 = 0 \Rightarrow x = 2$ and $x - 4 = 0 \Rightarrow x = 4$. The graph sho.

**19** **[Challenging]** • **Target: Solving Complex Equations** • **Cross-Lesson: 1.04 Partial Fractions**

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Determine the sum of the x-intercepts.

The x-intercepts of a function are the values of x for which $f(x) = 0$. For the quadratic function $f(x) = x^2 - 4x + 3$, we can find the x-intercepts by solving the equation $x^2 - 4x + 3 = 0$. This is a quadratic equation that can be factored. We are looking for two numbers that multiply to 3 and add to -4. These numbers are -1 and -3. Therefore, the equation can be factored as $(x-1)(x-3) = 0$. The solutions are $x = 1$ and $x = 3$. These are the x-intercepts. The sum of the x-intercepts is $1 + 3 = 4$. Alternative.

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**20** [Challenging] • Target: Argand Diagram Geometry & Loci • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.

1. First IBP: Let $u = x^2 \Rightarrow du = 2x\,dx$. Let $dv = \cos x\,dx \Rightarrow v = \sin x$. $\int_0^{\pi/2} x^2 \cos x\,dx = [x^2 \sin x]_0^{\pi/2} - 2\int_0^{\pi/2} x \sin x\,dx = \left(\left(\frac{\pi}{2}\right)^2(1) - 0\right) - 2\int_0^{\pi/2} x \sin x\,dx = \frac{\pi^2}{4} - 2J.$ 2. Second IBP on $J = \int_0^{\pi/2} x \sin x\,dx$: Let $u = x \Rightarrow du = dx$. Let $dv = \sin x\,dx \Rightarrow v = -\cos x$..

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Cartesian Arithmetic & Complex Conjugate

Given $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$, calculate $z_1 z_2$.

**a.** $5 + 14i$ ✓

**b.** $-3 - 8i$

**c.** $5 - 14i$

**d.** $-3 + 14i$

$z_1 z_2 = (3 - 2i)(-1 + 4i) = -3 + 12i + 2i - 8i^2 = -3 + 14i + 8 = 5 + 14i$.

**22** [Foundational] • Target: Cartesian Arithmetic & Complex Conjugate

Express $\frac{4 + 2i}{1 - i}$ in Cartesian form $a + bi$.

**a.** $1 + 3i$ ✓

**b.** $3 + 3i$

**c.** $1 - 3i$

**d.** $2 + 2i$

Multiply numerator and denominator by conjugate $(1 + i)$: $\frac{(4 + 2i)(1 + i)}{(1 - i)(1 + i)} = \frac{4 + 4i + 2i + 2i^2}{1 - i^2} = \frac{2 + 6i}{2} = 1 + 3i$.

**23** [Foundational] • Target: Cartesian Arithmetic & Complex Conjugate

If $z = x + iy$, what is $z \cdot z^*$ (where $z^*$ is the complex conjugate)?

**a.** $x^2 + y^2$ ✓

**b.** $x^2 - y^2$

**c.** $x^2 + 2ixy - y^2$

**d.** $2x$

$zz^* = (x + iy)(x - iy) = x^2 - (iy)^2 = x^2 - i^2 y^2 = x^2 + y^2 = |z|^2$.

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**24** [Foundational] • Target: Modulus and Argument

Find the modulus $|z|$ and principal argument $\text{Arg}(z) \in (-\pi, \pi]$ for $z = -\sqrt{3} + i$.

**a.** $|z| = 2$, $\text{Arg}(z) = \frac{5\pi}{6}$ ✓

**b.** $|z| = 2$, $\text{Arg}(z) = -\frac{\pi}{6}$

**c.** $|z| = 4$, $\text{Arg}(z) = \frac{2\pi}{3}$

**d.** $|z| = 2$, $\text{Arg}(z) = \frac{7\pi}{6}$

$|z| = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = 2$. In Quadrant II ($x < 0, y > 0$), reference angle is $\arctan(1/\sqrt{3}) = \frac{\pi}{6}$. Thus $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

**25** [Foundational] • Target: Modulus and Argument

What is the modulus of $\frac{3+4i}{1-2i}$?

**a.** $\sqrt{5}$ ✓

**b.** $\frac{5}{3}$

**c.** $5$

**d.** $\frac{25}{5} = 5$

Using modulus properties: $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|} = \frac{\sqrt{3^2+4^2}}{\sqrt{1^2+(-2)^2}} = \frac{5}{\sqrt{5}} = \sqrt{5}$.

**26** [Foundational] • Target: Modulus and Argument

Find the principal argument of $z = -5i$.

**a.** $-\frac{\pi}{2}$ ✓

**b.** $\frac{3\pi}{2}$

**c.** $\frac{\pi}{2}$

**d.** $\pi$

The point lies on the negative imaginary axis, which corresponds to angle $-\frac{\pi}{2}$ within the principal range $(-\pi, \pi]$.

**27** [Foundational] • Target: Solving Complex Equations

Solve the equation $2z + 3z^* = 15 - 2i$ for $z = x + iy$.

**a.** $z = 3 + 2i$ ✓

**b.** $z = 3 - 2i$

**c.** $z = 5 - 2i$

**d.** $z = 2 + 3i$

$2(x+iy) + 3(x-iy) = 5x - iy = 15 - 2i$. Equating real parts: $5x = 15 \Rightarrow x = 3$. Imaginary parts: $-y = -2 \Rightarrow y = 2$. Thus $z = 3 + 2i$.

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**28** [Foundational] • Target: Solving Complex Equations

Find the two square roots of $-8-6i$ in Cartesian form.

**a.** $\pm(1-3i)$ ✓

**b.** $\pm(3-i)$

**c.** $\pm(1+3i)$

**d.** $\pm(2-3i)$

Let $(x+iy)^2 = x^2 - y^2 + 2ixy = -8-6i$. Thus $x^2 - y^2 = -8$ and $2xy = -6 \Rightarrow y = -3/x$. $x^2 - 9/x^2 = -8 \Rightarrow x^4 + 8x^2 - 9 = 0 \Rightarrow (x^2+9)(x^2-1) = 0$. Since $x \in \mathbb{R}$, $x = \pm 1$, giving $y = \mp 3$. Roots are $\pm(1-3i)$.

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Solving Complex Equations

Solve the quadratic equation $z^2 - 4z + 13 = 0$ over $\mathbb{C}$.

**a.** $z = 2 \pm 3i$ ✓

**b.** $z = 4 \pm 6i$

**c.** $z = -2 \pm 3i$

**d.** $z = 2 \pm 9i$

By quadratic formula: $z = \frac{4 \pm \sqrt{16-52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i$.

**30** [Intermediate] • Target: Argand Diagram Geometry & Loci

Describe the locus of points in the complex plane satisfying $|z - 3 + 2i| = 5$.

**a.** **A circle with center** $(3, -2)$ **and radius** $5$ ✓

**b.** A circle with center $(-3, 2)$ and radius $5$

**c.** A straight line perpendicular to $3 - 2i$

**d.** An ellipse with focal distance $5$

Rewrite as $|z - (3 - 2i)| = 5$. The distance from $z$ to the fixed point $3 - 2i$ is constant and equal to $5$, which defines a circle centered at $(3, -2)$ with radius $5$.

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**31** [Intermediate] • Target: Argand Diagram Geometry & Loci

What geometric locus is defined by $|z-1| = |z-3i|$?

**a.** **The perpendicular bisector of the segment connecting $(1, 0)$ and $(0, 3)$** ✓

**b.** A circle passing through $(1, 0)$ and $(0, 3)$

**c.** A line segment connecting $(1, 0)$ and $(0, 3)$

**d.** A parabola opening upward

$|z - z_1| = |z - z_2|$ represents the set of all points equidistant from $z_1 = 1 + 0i$ and $z_2 = 0 + 3i$, which is the perpendicular bisector of the segment connecting them.

**32** [Intermediate] • Target: Argand Diagram Geometry & Loci

Find the Cartesian equation of the locus $\text{Arg}(z-2) = \frac{\pi}{4}$.

**a.** **The half-line $y = x - 2$ for $x > 2$** ✓

**b.** The full line $y = x - 2$

**c.** The circle $(x-2)^2 + y^2 = 1$

**d.** The half-line $y = -x + 2$ for $x < 2$

$\text{Arg}((x-2)+iy) = \frac{\pi}{4} \Rightarrow \frac{y}{x-2} = \tan(\pi/4) = 1 \Rightarrow y = x - 2$, with the ray restriction $x - 2 > 0 \Rightarrow x > 2, y > 0$.

**33** [Intermediate] • Target: Cartesian Arithmetic & Complex Conjugate

How many distinct complex $n$th roots does the equation $z^7 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

**a.** $7$ roots, $\frac{360}{7}^\circ$ apart ✓

**b.** $8$ roots, $\frac{360}{7}^\circ$ apart

**c.** $7$ roots, $45^\circ$ apart

**d.** $6$ roots, $\frac{360}{7}^\circ$ apart

$z^7 = 1$ has exactly $7$ roots, equally spaced by $\frac{360^\circ}{7} = \frac{360}{7}^\circ$.

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**34** [Intermediate] • Target: Modulus and Argument

How many distinct complex $n$th roots does the equation $z^3 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

**a.** 3 roots, $90^\circ$ apart

**b.** 2 roots, $120^\circ$ apart

**c.** 3 roots, $120^\circ$ apart ✓

**d.** 4 roots, $120^\circ$ apart

$z^3 = 1$ has exactly 3 roots, equally spaced by $\frac{360^\circ}{3} = 120^\circ$.

**35** [Intermediate] • Target: Solving Complex Equations

How many distinct complex $n$th roots does the equation $z^5 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

**a.** 5 roots, $72^\circ$ apart ✓

**b.** 4 roots, $72^\circ$ apart

**c.** 5 roots, $60^\circ$ apart

**d.** 6 roots, $72^\circ$ apart

$z^5 = 1$ has exactly 5 roots, equally spaced by $\frac{360^\circ}{5} = 72^\circ$.

**36** [Intermediate] • Target: Argand Diagram Geometry & Loci

How many distinct complex $n$th roots does the equation $z^4 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

**a.** 4 roots, $72^\circ$ apart

**b.** 4 roots, $90^\circ$ apart ✓

**c.** 5 roots, $90^\circ$ apart

**d.** 3 roots, $90^\circ$ apart

$z^4 = 1$ has exactly 4 roots, equally spaced by $\frac{360^\circ}{4} = 90^\circ$.

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**37** [Intermediate] • Target: Cartesian Arithmetic & Complex Conjugate

How many distinct complex $n$th roots does the equation $z^8 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

**a.** 9 roots, $45^\circ$ apart

**b.** 8 roots, $40^\circ$ apart

**c.** 7 roots, $45^\circ$ apart

**d.** 8 roots, $45^\circ$ apart ✓

$z^8 = 1$ has exactly 8 roots, equally spaced by $\frac{360^\circ}{8} = 45^\circ$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Modulus and Argument • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Consider the complex number $w = 4 - 3i$. Find the modulus of $w$, denoted as $|w|$.

**a.** **5** ✓

**b.** 6

**c.** 7

**d.** 25

The modulus of a complex number $z = x + yi$ is given by $|z| = \sqrt{x^2 + y^2}$. For the complex number $w = 4 - 3i$, we have $x = 4$ and $y = -3$. Therefore, $|w| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$. Common errors include: - Forgetting to square the imaginary part: $\sqrt{4 + (-3)} = \sqrt{1} = 1$ (Incorrect). - Forgetting to take the square root: $4^2 + (-3)^2 = 16 + 9 = 25$ (Incorrect, as this is the square of the modulus). - Incorrectly handling the negative sign in the imaginary part.

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**39** [Challenging] • Target: Solving Complex Equations • Cross-Lesson: 1.04 Partial Fractions

Let $z = x + yi$ be a complex number. The modulus of $z$, denoted by $|z|$, is given by $|z| = \sqrt{x^2 + y^2}$. Given that $|z| = \sqrt{5}$ and the real part of $z$ is $2$, find the possible values for the imaginary part of $z$.

**a.** $\pm\sqrt{3}$ ✓

**b.** $\pm 3$

**c.** $\pm\sqrt{1}$

**d.** $\pm 1$

We are given that $z = x + yi$. The modulus of $z$ is $|z| = \sqrt{x^2 + y^2}$. We are given that $|z| = \sqrt{5}$, so $\sqrt{x^2 + y^2} = \sqrt{5}$. Squaring both sides, we get $x^2 + y^2 = 5$. We are also given that the real part of $z$ is $2$, which means $x = 2$. Substituting $x = 2$ into the equation $x^2 + y^2 = 5$, we get $2^2 + y^2 = 5$, which simplifies to $4 + y^2 = 5$. Subtracting 4 from both sides, we have $y^2 = 1$. Taking the square root of both sides, we find $y = \pm 1$. Therefore, the possible valu.

**40** [Challenging] • Target: Argand Diagram Geometry & Loci • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Consider the complex number $z = 3 + 4i$. What is the modulus of $z$?

**a.** **5** ✓

**b.** 7

**c.** 25

**d.** $\sqrt{7}$

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For the given complex number $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the modulus of $z$ is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Common misconceptions: - Adding the real and imaginary parts: $3 + 4 = 7$, which is incorrect. - Squaring the real and imaginary parts without taking the square root: $3^2 + 4^2 = 9 + 16 = 25$, which is the square of the modulus. - Incorrectly calculating the square root.

**41** [Challenging] • Target: Cartesian Arithmetic & Complex Conjugate • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Consider the complex number $z = 3 + ki$, where $k$ is a real parameter. The modulus of $z$ is given by $|z| = \sqrt{13}$. Find the positive value of $k$.

**a.** **1** ✓

**b.** 2

**c.** 3

**d.** 4

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For the complex number $z = 3 + ki$, we have $a = 3$ and $b = k$. Therefore, $|z| = \sqrt{3^2 + k^2} = \sqrt{9 + k^2}$. We are given that $|z| = \sqrt{13}$. So, we have the equation: $\sqrt{9 + k^2} = \sqrt{13}$ Squaring both sides of the equation, we get: $9 + k^2 = 13$ Subtracting 9 from both sides: $k^2 = 13 - 9$ $k^2 = 4$ Taking the square root of both sides, we get $k = \pm 2$. Since we are asked for the positive value of $k$, w.

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**42** [Challenging] • Target: Modulus and Argument • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Consider the complex number $z = -3 + 4i$. Find the modulus of $z$, denoted by $|z|$.

**a.** **5** ✓

**b.** 7

**c.** 25

**d.** 16

The modulus of a complex number $z = a + bi$ is given by the formula $|z| = \sqrt{a^2 + b^2}$. In this case, $a = -3$ and $b = 4$. Therefore, $|z| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

**43** [Challenging] • Target: Solving Complex Equations • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Write $z = 5(\cos 135^\circ + i\sin 135^\circ)$ in the form $x + yi$ (exact values).

**a.** $-\frac{5\sqrt{2}}{2} - \frac{5\sqrt{2}}{2}i$

**b.** $\frac{5\sqrt{2}}{2} - \frac{5\sqrt{2}}{2}i$

**c.** $\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$

**d.** $-\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$ ✓

Working through this complex-numbers step: $z = 5\cos 135^\circ + 5\sin 135^\circ\, i = -\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$

**44** [Challenging] • Target: Argand Diagram Geometry & Loci • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Write $z = 3(\cos 135^\circ + i\sin 135^\circ)$ in the form $x + yi$ (exact values).

**a.** $-\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$ ✓

**b.** $-\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i$

**c.** $\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i$

**d.** $\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$

Working through this complex-numbers step: $z = 3\cos 135^\circ + 3\sin 135^\circ\, i = -\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$

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**45** [Challenging] • Target: Cartesian Arithmetic & Complex Conjugate • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Write $z = 3(\cos 150^\circ + i\sin 150^\circ)$ in the form $x + yi$ (exact values).

**a.** $-\frac{3\sqrt{3}}{2} + \frac{3}{2}i$ ✓

**b.** $\frac{3}{2} - \frac{3\sqrt{3}}{2}i$

**c.** $\frac{3\sqrt{3}}{2} + \frac{3}{2}i$

**d.** $-\frac{3\sqrt{3}}{2} - \frac{3}{2}i$

Working through this complex-numbers step: $z = 3\cos 150^\circ + 3\sin 150^\circ\, i = -\frac{3\sqrt{3}}{2} + \frac{3}{2}i$

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